

Brazil's Economic Development Agenda

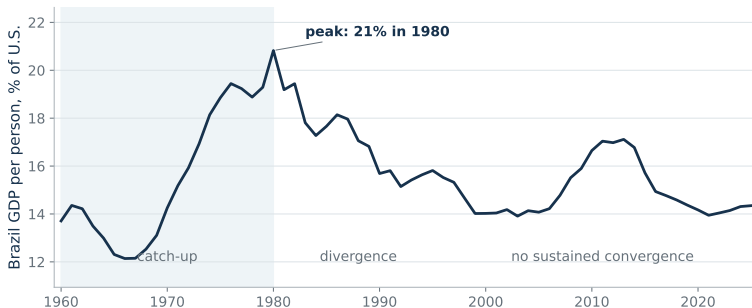
Macroeconomic stability, productivity, state capacity, and inclusion

Eduardo Peres Cunha and Arthur Lins de Vasconcellos

PUC-Rio – Brazilian Seminars | July 2026

Data cutoff: July 2026

Relative to U.S. income, Brazil is where it was in 1960



Brazil converged rapidly until 1980. Why did it stop, and what would make it resume?

Source: World Bank, World Development Indicators, GDP per capita in constant 2015 US\$ (NY.GDP.PCAP.KD), 1960–2025.

Brazil's position raises two questions

Part I: Why are we like this?

How did macroeconomic instability, slavery, rapid urbanization, and natural resources shape institutions, inequality, and production?

Part II: Where do we go from here?

Which structural problems must Brazil solve, and which productive capabilities can it expand?

The development agenda begins with Brazil's history and ends with a choice about its capabilities.

Where is Brazil relative to other countries?

Countries can fall behind

Argentina and Venezuela show that earlier prosperity does not guarantee continued convergence.

Brazil remains in the middle

A large and diversified economy has not completed the transition to rich-country productivity and income.

Countries can move closer

Chile and South Korea illustrate different paths toward higher income and development.

Why has Brazil remained in the middle, and what would be required to complete the transition?

Source: World Bank, World Development Indicators. Country examples describe development trajectories, not formal income-group classifications.

Brazil is a continental-scale country



8.5
million
km²

- Scale raises the cost of connecting people, firms, and public services.
- Climate, geography, and settlement patterns differ sharply within the same country.

Source: IBGE, Territorial Areas 2025. The geographic overlay is illustrative.

A large economy with concentrated income

National scale, 2024

212.6 million people

US\$2.19 trillion GDP

- ▶ More people than Germany, France, and Spain combined.
- ▶ An economy roughly the size of Canada's.

Gini Index, 2024

0.506

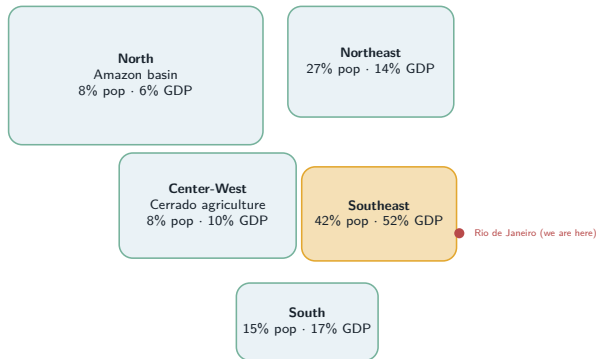
0 means equal incomes; 1 means one person receives all income.

For comparison: France's is 0.30 and the U.S.'s is 0.49

Brazil has the scale to build capabilities, but access to them remains unequal.

Source: IBGE, Population Estimates and PNAD Continuous 2024; World Bank, WDI.

One country, five very different economies



Schematic: positions indicative, areas not to scale.

Source: IBGE, 2022 Census and Regional Accounts 2021. Values rounded.

- ▶ **Southeast:** 42% of the population and 52% of GDP.
- ▶ **North and Northeast:** 35% of the population and 20% of GDP.
- ▶ **Center-West:** 10% of GDP and a major agricultural frontier.
- ▶ National policies meet different infrastructure, demographics, and local state capacity.

What exactly is economic development?

- ▶ **Income:** a lasting rise in material living standards.
- ▶ **Productivity:** more output from each worker and unit of capital.
- ▶ **Inclusion:** broader access to education, formal work, services, and mobility.
- ▶ **Resilience:** institutions and infrastructure that withstand fiscal, external, and climate shocks.

The lecture follows two questions

Why are we like this?

- ▶ Institutions and macroeconomic stability
- ▶ Historical formation
- ▶ Inequality and productive structure
- ▶ Brazil today

Where do we go from here?

- ▶ Problems to solve
- ▶ Productive opportunities
- ▶ Political economy and implementation
- ▶ An integrated development agenda

Part I – Why are we like this?

History, institutions, inequality, and production

Four historical forces still shape Brazil today

Macroeconomic instability

Inflation and debt crises shortened planning horizons and weakened public investment.

Slavery

Forced labor and exclusion left persistent racial gaps in assets, education, and political power.

Rapid urbanization

Cities expanded faster than housing, transport, sanitation, and local state capacity.

Natural resources

Land, energy, and minerals created global advantages and recurring governance conflicts.

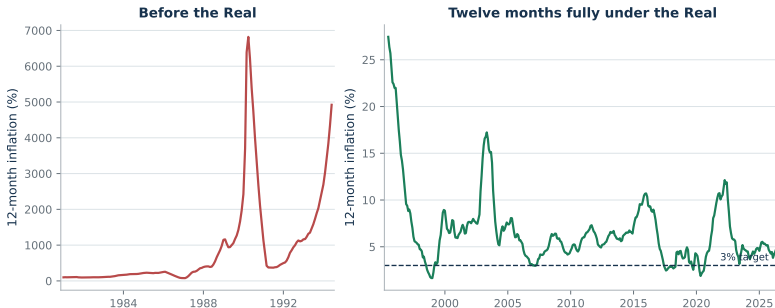
Historical formation shaped Brazil's institutions, inequality, and productive structure.

PART I

Macroeconomic stability

How Brazil stopped chronic inflation — and why stabilization did not guarantee convergence

Before 1994, inflation made the economy unstable



- Prices changed faster than wages, contracts, and public budgets could adjust; twelve-month inflation reached thousands of percent.
- Stabilization means a durable break from this regime, not the absence of every later inflation spike.

Source: BCB SGS 433, monthly IPCA. Rates are compounded over twelve months. The two panels use different vertical scales.

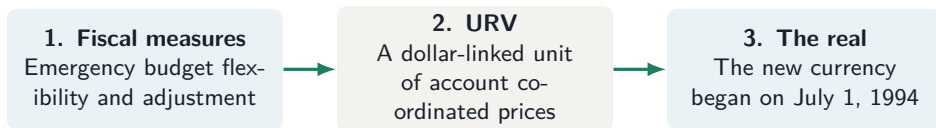
Stabilization restored a usable price system



- ▶ Stable prices restored information and lengthened the horizon for household, firm, and government decisions.
- ▶ Later spikes show that price stability still depends on credible fiscal and monetary institutions.

Source: BCB SGS 433, monthly IPCA. Twelve-month rates are compounded from the official monthly series.

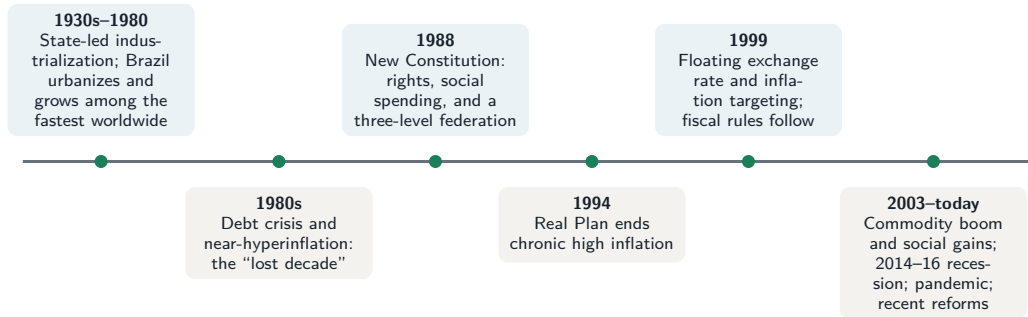
How did Brazil stabilize? The Real Plan



- ▶ The gradual conversion broke inflation inertia without a surprise price freeze.
- ▶ Former PUC-Rio faculty played a central role in developing and implementing the plan.

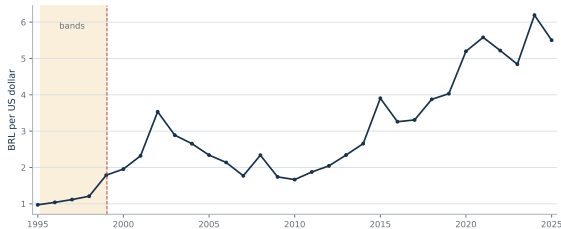
Source: BCB, [The Real Plan](#); PUC-Rio Department of Economics, institutional history.

Six moments place stabilization in a longer history



- Stabilization is one turn in a longer institutional history that still shapes Brazil's constraints and capabilities.

1999: the exchange-rate anchor gave way to a float



1994–1998: defend the anchor

The exchange rate anchored tradable prices; interest rates and reserves defended the band.

1999 onward: let the currency adjust

Reserve losses forced a float. The inflation target became the anchor, and the currency absorbed more shocks.

Source: BCB SGS 3692; BCB, 1999 FX Market Analysis and Foreign Exchange Policy.

The exchange rate absorbs shocks and changes domestic prices

Why it is useful

A floating currency changes relative prices when commodity prices, foreign interest rates, or capital flows move.

Why it is volatile

The real reacts quickly to the global dollar, interest differentials, risk, fiscal news, and trade flows.

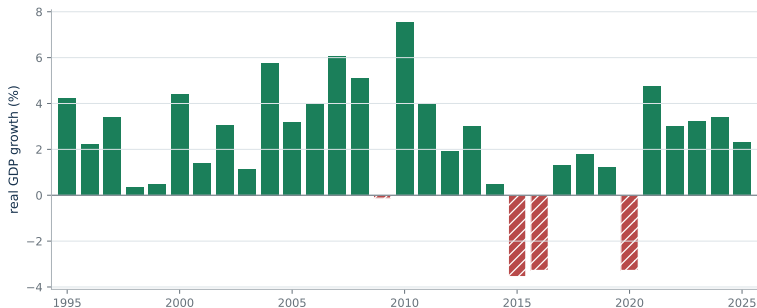
Why it matters

It affects imported inflation, export margins, dollar debts, financial conditions, and the allocation of production.

The exchange rate is both a shock absorber and a channel through which shocks reach the economy.

Source: BCB, exchange-rate policy and monetary-policy transmission channels.

Growth after stabilization has remained volatile



- Brazil combined strong expansions with three recessions after 1995; average growth weakened before the pandemic.
- The macro regime reduced one source of crisis, but the convergence line still failed to trend upward.

Source: BCB SGS 7326, based on IBGE National Accounts. Annual real GDP growth; 2025 is the latest observation. Hatched bars mark contractions.

Brazil's fiscal problem is about composition and credibility

The state is not small

Revenue and social spending are substantial for an upper-middle-income economy.

Much spending is rigid

Pensions, payroll, transfers, and earmarks limit annual reallocation.

Development margin

Public investment fell to roughly 2% of GDP — among the lowest shares in the G20 — as rigid current spending grew.

Macro margin

Uncertain debt paths raise risk premia and shorten private planning horizons.

Source: OECD, [Economic Survey: Brazil 2023](#).

Debt dynamics compress the space for development spending



- ▶ Debt rose before the pandemic, jumped in 2020, and remains high relative to the mid-2000s.
- ▶ Interest rates, primary balances, growth, and debt composition jointly determine the path.

Source: BCB SGS 13762. Gross general government debt, methodology used since 2008, monthly, percent of GDP.

Low saving and investment constrain capital deepening

16.8%
investment / GDP, 2025

14.4%
saving / GDP, 2025

- ▶ Capital per worker grows slowly when investment remains low.
- ▶ High financing costs screen out long-horizon projects.
- ▶ Infrastructure bottlenecks reduce the return to private capital.
- ▶ Productivity growth matters because more investment alone can misallocate resources.

Source: IBGE, National Accounts 2025; rates are shares of GDP.

Macro synthesis: three messages

1. **The Real Plan was an institutional solution**, not just a new currency: the URV coordinated expectations, and fiscal measures backed it.
2. **1999 changed who absorbs shocks**: the float, the inflation target, fiscal rules, and reserves work as one architecture.
3. **Stability is not convergence**: debt uncertainty, rigid budgets, and a high cost of capital still shorten horizons and compress investment.

Stable prices solved one historical constraint. Inequality, weak diffusion, and uneven state capacity remained.

Source: Full macro scorecard table in the appendix.

PART I

Historical formation

Slavery, urbanization, racial segregation, and the natural-resource base

Slavery deeply shaped the Brazilian economy and society

About 4 million

Africans were forced to Brazil between the sixteenth and mid-nineteenth centuries. The most in the whole world.

15.2% in 1872

The first national census counted 1.51 million enslaved people.

1888

Brazil was the last country in the Americas to abolish slavery.

- The two-article Lei Aurea ended legal slavery but set no path for land, work, or education.

Source: IBGE, *Brazil: 500 Years of Settlement* and 1872 Census; Federal Senate, Law 3,353/1888.

Race and territory overlap inside Brazilian cities

Brazil overall

55.5%

Black or brown

Favelas and urban communities

72.9%

Black or brown

- ▶ Black Brazilians were 10.2% of the population but 16.1% of favela residents; brown Brazilians were 45.3% and 56.8%, respectively.
- ▶ Spatial segregation concentrates exposure to weak infrastructure and raises the cost of reaching opportunity.

Source: IBGE, 2022 Census, Favelas and Urban Communities, Table 9884.

Racial gaps persist in skills, earnings, and basic services

Illiteracy, 2022

White: **4.3%**
Black: **10.1%**
Brown: **8.8%**

Hourly earnings, 2024

White: **R\$24.60**
Black or brown: **R\$15.00**

Adequate sewage, 2022

White: **83.5%**
Black: **75.0%**
Brown: **68.9%**

- Territory, education, occupation, informality, networks, and discrimination reinforce one another over the life cycle.

Source: IBGE, 2022 Census, Literacy and Household Characteristics; Social Indicators Synthesis 2025, PNAD Continua 2024.

Brazil became an urban country in one generation

36.2%
urban population, 1950

55.9%
urban population, 1970

87.4%
urban population, 2022

- ▶ Industrialization and rural migration moved the development problem toward cities within a few decades.
- ▶ Housing, transport, sanitation, schools, and municipal capacity had to expand at the same speed.

Source: IBGE, Demographic Censuses 1950, 1970, and 2022. Urban shares use the official legal classification.

Urban growth outpaced formal housing and infrastructure

16.4m
residents

8.1%
of Brazil's population

12,348
favelas and urban
communities

- ▶ Informal settlements were a household response to scarce serviced land and insufficient housing policy.
- ▶ The result is not only a housing deficit: tenure, drainage, transport, public space, and service delivery interact to shape opportunity and inequality.

Source: IBGE, 2022 Census, Favelas and Urban Communities. Revised universe results: 16,390,790 residents.

Where people live determines which opportunities they can reach

56.8%

took 6–30 minutes to reach work

10.5%

took more than one and up to two hours

1.8%

took more than two hours — about 1.3 million workers

- ▶ Trips above two hours reached 1.4% among white workers, 2.5% among Black workers, and 2.0% among brown workers.

Segregation is economic: it changes job matching, school choice, time use, and firms' access to workers.

Source: IBGE, 2022 Census, preliminary sample results on commuting to work.

Half of Brazil's total energy supply is renewable

50.0%
renewables in total energy
supply

322.0
million tonnes of oil
equivalent

+2.4%
energy supply growth in 2024

- ▶ Hydropower and sugarcane biomass created the base; wind, solar, black liquor, and biodiesel raised the renewable share.
- ▶ Petroleum and derivatives still supplied 109.6 million tonnes of oil equivalent, so transport and heat remain central.

Source: EPE, Brazilian Energy Balance 2025, base year 2024.

Electricity is cleaner — and wind and solar are diversifying it

88.2%
renewable electricity supply

23.7%
wind and solar generation

59.9
kg CO₂e per MWh

- ▶ Hydropower generated 421.8 TWh in 2024; wind generated 107.7 TWh and solar 70.7 TWh.
- ▶ A diversified clean grid can support low-carbon industry, but it requires transmission, storage, and predictable licensing.

Source: EPE, Brazilian Energy Balance 2025, base year 2024.

Mineral wealth is large — and highly concentrated

R\$220.5bn

metallic minerals and
graphite

R\$159bn

iron ore alone

US\$42bn

trade surplus in 15 minerals

- ▶ Iron ore accounted for 72% of the measured production value; Minas Gerais and Pará produced 96.7% of beneficiated iron ore.

Natural resources create rents and exports. Infrastructure, suppliers, skills, safeguards, and fiscal rules turn them into development.

Source: ANM, Brazilian Mineral Yearbook 2025 and Mineral Summary 2025, data for 2024.

Short break

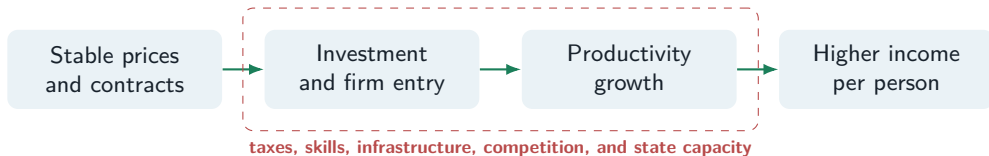
After the break: how historical formation appears in Brazil today

PART I

Brazil today

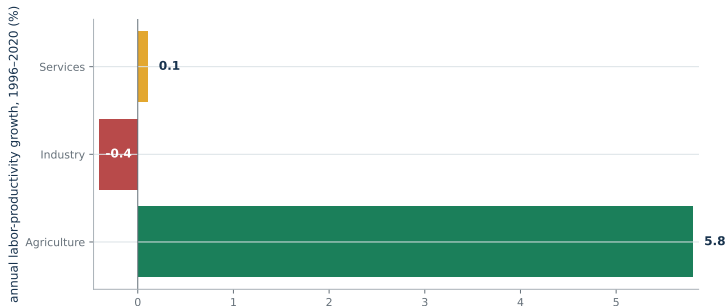
Institutions, inequality, production, and state capacity

Stability raises the return to investment; it does not guarantee it



Brazil solved much of the first arrow. The middle of the chain remains weak.

Brazil's productivity problem is not uniform across sectors



- ▶ The bars are average annual growth rates of *labor productivity* by sector, 1996–2020.
- ▶ Agriculture shows that sustained productivity growth is possible; services employ most workers, so diffusion beyond the frontier sector determines the aggregate result.

Source: World Bank, labor-productivity growth by sector, 1996–2020; values rounded.

Low unemployment can coexist with a dual labor market



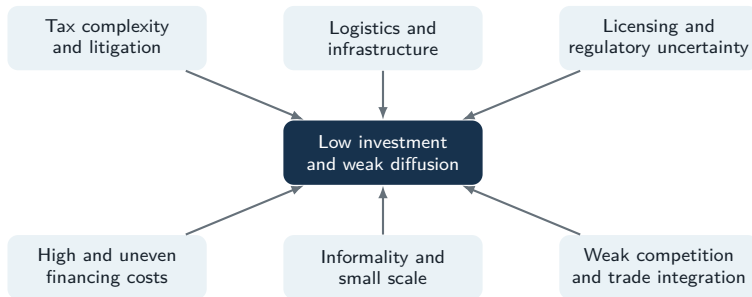
Informality

38.1%
of workers in 2025

- insurance: income and flexibility
- trap: no credit, scale, or training
- policy: lower formalization cliffs, portable benefits

Source: BCB SGS 24369, sourced from IBGE PNAD Contínua; IBGE annual informality rate for 2025. Informality is a statistical labor-market category, not a synonym for tax evasion.

Custo Brasil: a system of interacting wedges



Removing one wedge may have little effect when the other complements remain binding.

We focus next on three of these wedges — infrastructure, credit, and skills; taxes, competition, and trade are covered in the appendix.

Infrastructure gaps: logistics and sanitation

Logistics

- ▶ A continental economy makes transport time and reliability part of marginal cost.
- ▶ Roads, rail, ports, storage, and digital systems are complements.

Sanitation

- ▶ Water and sewage coverage affect disease, school attendance, time use, and property values.
- ▶ National averages conceal large regional and within-city gaps.

A new asset raises productivity only if governance, maintenance, and the network around it work.

Source: World Bank, Brazil infrastructure diagnostics; EPL/Infra S.A., National Logistics Plan; SNIS, Ministry of Cities.

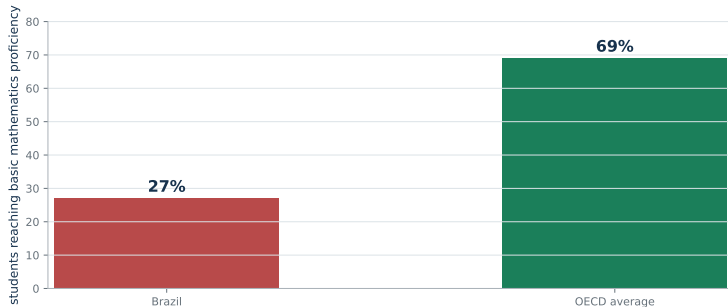
The credit spread remains a microeconomic wedge



- Concretely: with a policy rate near 10% and a 20-point spread, a small firm borrows at about 30% a year — a machine must return that much before financing it makes sense.
- The spread combines expected losses, operating costs, taxes, market power, and risk premia; collateral rules, information sharing, competition, and insolvency reform address different parts.

Source: BCB SGS 20783. Average spread on outstanding credit, monthly, percentage points.

The human-capital constraint is the quality of education system



- ▶ PISA tests 15-year-old students; “basic proficiency” is Level 2 in mathematics — using simple procedures to solve everyday problems.
- ▶ Learning gaps reduce technology adoption and worker mobility.

Source: OECD, [PISA 2022: Brazil](#).

Inequality is racial, regional, and gendered

Race and income

White workers earn about 70% more on average; Black and brown workers are overrepresented in informal jobs.

Region and services

Sanitation, transport, and school gaps concentrate in majority-Black peripheries and the North and Northeast.

Gender and care

Women carry more unpaid care work and face lower labor-force participation, especially when childcare is scarce.

- ▶ Black and brown Brazilians are 55.5% of the population. Race, region, and gender overlap in access to safe neighborhoods, schools, formal work, and political voice.

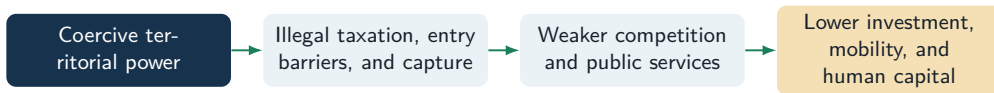
These are not separate topics: they determine who can build skills, move, invest, and participate.

Source: IBGE, 2022 Census, Desigualdades Sociais por Cor ou Raça no Brasil, and PNAD Contínua care and labor-force indicators; Ipea, Poverty and Inequality 1995–2024.

Case study – Security and development

How coercive territorial power changes markets, services, and state capacity

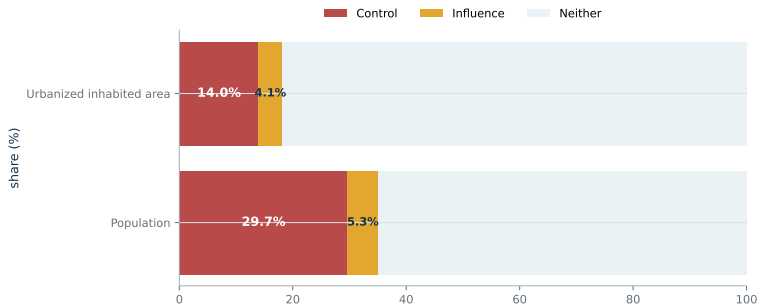
Organized crime is a microeconomic constraint



- ▶ Extortion acts like an illegal tax; forced suppliers create local market power over gas, internet, and transport.
- ▶ Land appropriation weakens property rights and collateral.
- ▶ Intimidation and corruption can redirect enforcement and public resources.

Source: MPRJ documented cases; SENAPPEN; GENI/UFF and Instituto Fogo Cruzado. Mechanisms are not estimates of aggregate GDP losses. Measurement details in the appendix.

Rio de Janeiro: Criminal Governance



- ▶ In 2024, 34.9% of residents were estimated to live under armed-group control or influence.
- ▶ That's over 4 million people, more than the population of 12 Brazilian states.

Source: GENI/UFF and Instituto Fogo Cruzado, Mapa Histórico dos Grupos Armados 2025, with 2022 Census population data. Published shares are rounded.

Territorial violence erodes human capital

Exposure

51%

of students in the 19 municipalities studied were affected by armed violence in 2022.

Learning

10 points

maximum reported educational gap, interpreted by the authors as about six months of learning in average.

Dropout

**7.2 vs.
9.0–9.6%**

non-controlled versus controlled areas in Metropolitan Rio.

Source: UNICEF, Educação Sob Cerco 2025, with GENI/UFF and Instituto Fogo Cruzado. Conditional estimates with statistical controls; the design is not experimental.

A credible response combines enforcement and service delivery

Protect territory and services

Safe routes, reliable transport, service continuity, youth prevention.

Investigate and prosecute

Forensics, witness protection, case management, corruption control.

Follow the money

Beneficial ownership, financial intelligence, freezing and confiscation.

Source: MJSP, ENFOC and ENCCLA; SENAPPEN/CNPCP, National Plan 2024–2027. Full instrument-outcome table and discussion exercise in the appendix.

Part II – Where do we go from here?

Problems to solve and opportunities to exploit

Four structural problems limit Brazil's development

Organized crime

Territorial control weakens the state, public services, property rights, and market entry.

Fiscal challenges

Pension (INSS) obligations, rigid spending, high debt, and interest costs compress investment and policy space.

Education

Enrollment has expanded, but weak learning limits productivity and social mobility.

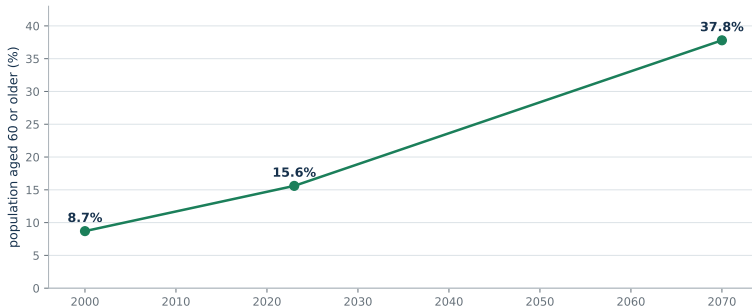
Innovation

Research, firms, finance, competition, and diffusion remain weakly connected outside selected sectors.

The Rio case study you just saw is the first of these problems in action.

Brazil must solve these problems while demography and climate change narrow the time available.

Demography raises the return to productivity now



- ▶ The working-age share will no longer provide the same contribution to growth.
- ▶ Health, pensions, care, lifelong learning, and later-life employment become linked policies.

Source: IBGE, Population Projections 2024. The 2070 value is a projection, not an observed outcome.

Three topics affect the whole outlook

Demography

Population peaks around 2041. The 60+ share rises from 15.6% in 2023 to 37.8% in 2070.

Climate

Droughts, floods, heat, and land-use pressure affect capital, food, energy, and cities.

Fiscal space

Aging raises health and pension pressure while adaptation and infrastructure require investment.

Future growth must rely more on productivity because labor-force expansion is ending.

Source: IBGE, Population Projections 2024; OECD, Economic Survey Brazil 2023.

Implementation is part of every reform

The federal bargain

One Union, 27 states, and 5,570 municipalities share taxes and mandates; major reforms need constitutional amendments.

A rigid budget

Earmarks and mandatory spending leave little room to reallocate each year.

Concentrated losers, diffuse winners

Sectors with special regimes organize and litigate; winning consumers are dispersed. The 2023 tax reform passed by compensating losers over decades.

Unequal implementation

A law is only as good as the local administration that applies it.

A reform's design includes its coalition, compensation, and sequencing.

Opportunities

Existing strengths can support new industries, exports, and higher-productivity services

Brazil can expand on four sectors

Clean energy

Renewable electricity, biofuels, and low-carbon fuels can attract production and support new exports.

Agribusiness

Research and tropical agriculture can extend into logistics, technology, processing, and higher-value products.

Knowledge centers

São Paulo and Rio combine universities, researchers, firms, capital, and specialized workers.

Tourism

Natural, cultural, and urban attractions remain constrained by safety, connectivity, and local public services.

The opportunity is not a blank slate: it starts from capabilities Brazil already has.

Brazil's energy scenario is a global advantage

Energy advantage

88.2%

renewable domestic electricity supply in 2024

- ▶ Cheap clean power is an export and investment pitch: low-carbon industry, fuels, and agriculture.
- ▶ Climate runs through every earlier block: exports and the exchange rate, infrastructure and adaptation, agriculture, urban risk, and the budget.

Emissions structure

about 70%

from land use and agriculture in the 2022 inventory

- ▶ Enforcement, livestock productivity, and restoration are simple to implement.

Source: EPE, Brazilian Energy Balance 2025, base year 2024; MCTI, National Emissions Inventory, 2022 results.

Agribusiness is already global

Weight in the
economy

23%

of Brazil's GDP (2024)

Weight in trade

49%

of exports – US\$164 bn
(2024)

Record harvest

350 Mt

of grain in 2024/25

- ▶ **World's no. 1 exporter** of soybeans, coffee, sugar, orange juice, beef and chicken – built on Embrapa's tropical-agriculture science, which turned the Cerrado into farmland.
- ▶ The next step is **robustness** – processing, agtech, biologicals, machinery, logistics – and credibility abroad rests on cutting **logistics costs** and enforcing **environmental rules**.

Feeding the world is solved; capturing more value per ton is not.

Source: CEPEA/CNA agribusiness GDP (2024); Ministry of Agriculture trade data (2024); CONAB 12th Harvest Survey, 2024/25 season.

The knowledge ingredients exist; the links are weak

Universities and research

about half

of Latin America's scientific papers are Brazilian; USP tops the QS regional ranking

Firms and capital

#1

São Paulo is Latin America's largest startup hub;
~two-thirds of the region's unicorns are Brazilian

The missing link

1.2%

of GDP goes to research, versus **2.7%** in the OECD

- ▶ Rio has an energy research cluster, creative industries, and PUC-Rio — where you are sitting.
- ▶ Yet the pieces rarely connect: few researchers work in companies, so business research spending stays low.

Connecting frontier research and firms is a mission that aims to create new industries and raise productivity.

Tourism: world-class assets, everyday constraints

Natural assets (WEF index)

5th

of 119 countries – **top tier**
(overall: only 26th)

Safety & security pillar

110th

of 119 – **near the bottom**

Foreign visitors, 2024

6.7M

a record – yet Mexico
received 45M

- ▶ Why it matters: tourism creates jobs across skill levels and brings income to the Northeast and the interior – exactly where we saw the largest income gaps.
- ▶ The blockers are not tourism-specific: safety perception, costly flights, weak local services.

The product exists; the missing conditions are Brazil's general ones.

Source: WEF Travel & Tourism Development Index 2024 (pillar ranks computed from published scores, 119 economies); Embratur / Ministry of Tourism, 2024 arrivals; INEGI (Mexico), 2024.

Established capabilities can seed new global industries

Established global industries

- ▶ Mining — Vale
- ▶ Oil and energy — Petrobras
- ▶ Pulp and paper — Suzano and Klabin

Emerging opportunities

- ▶ Climate-related industries
- ▶ Digital products and services
- ▶ Financial technology — Pix

The development task is to connect existing firms, research, finance, workers, and suppliers.

Pix shows that Brazilian institutions can build at scale

170m+
individual Pix users

7bn+
transactions in January 2026

- ▶ The central bank created a free, instant-payment platform that reached national scale within a few years.
- ▶ Pix complements private banks and fintech firms; common infrastructure expands entry and innovation.

Source: BCB, [Pix Statistics](#); figures reported in early 2026.

The agenda joins structural problems to productive capabilities

Historical formation → Brazil today

Structural problems

Fiscal pressure, organized crime, weak learning, limited innovation, and unequal access to state capacity.

Development task

Solve fundamental institutional problems.

Productive capabilities

Clean energy, agriculture, knowledge cities, global firms, natural resources, and digital finance.

Development task

Expand and connect existing capabilities.

Both tasks lead to the same goal: a richer, more productive and more inclusive economy.

Brazil must solve constraints and expand capabilities together

What would be required to complete the transition?

Brazil already has substantial productive and institutional assets. The central challenge is to combine them with progress in fiscal policy, public security, education, and innovation.

Solve

Restore fiscal space, territorial control, learning, and the institutional links that turn research into innovation.

Expand

Use clean energy, productive agriculture, urban knowledge, global firms, tourism, and digital finance as platforms for diffusion.

Thank you!

Eduardo Peres Cunha and Arthur Lins de Vasconcellos
— PUC-Rio, Brazilian Seminars

Appendix

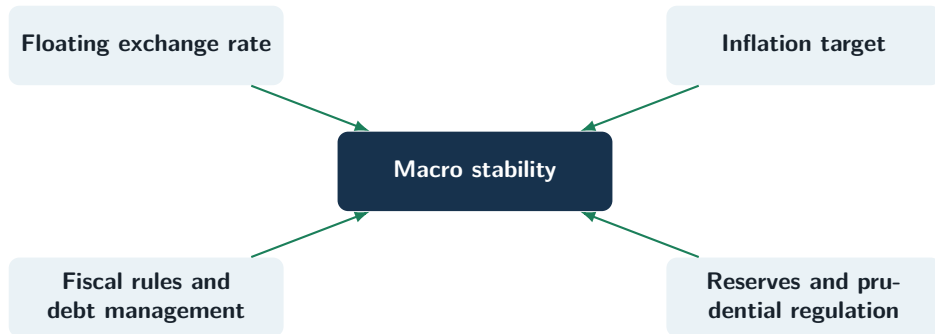
Backup slides: formulas, full tables, measurement, and references

Backup: six key terms

Income convergence	A poorer country grows faster than rich ones and closes the gap in income per person.
Nominal anchor	The variable a central bank commits to (an exchange rate, an inflation target) so that everyone's prices and wages can coordinate on it.
Pass-through	How much of a currency depreciation ends up in consumer prices.
Primary balance	Government revenue minus spending, <i>excluding</i> interest payments.
Reallocation	Workers and capital moving from less productive firms to more productive ones.
Contestability	Markets where new firms can credibly enter and challenge incumbents.

URV, CBS/IBS, and Selic are defined when they appear.

Backup: the post-1999 macro framework



- ▶ The pieces are complements: fiscal risk affects interest rates and the currency.
- ▶ The float absorbs shocks; monetary credibility limits pass-through.

Backup: nominal and real exchange rates answer different questions

Nominal rate

$$E = \text{BRL per USD}$$

A rise in E is a nominal depreciation of the real.

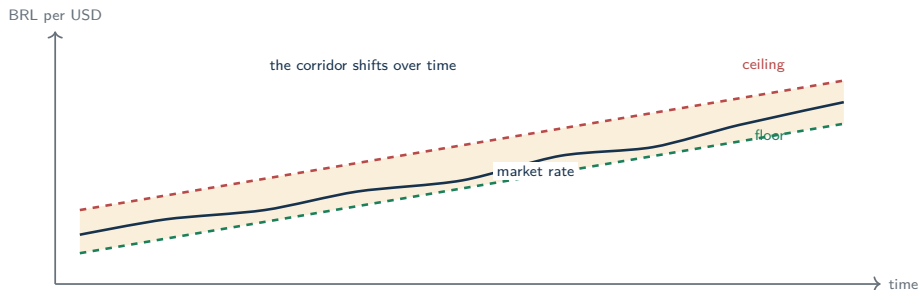
Real exchange rate

$$q = \frac{EP^*}{P}$$

A rise in q makes foreign goods more expensive relative to domestic goods under this definition.

- ▶ Domestic and foreign inflation can undo a nominal depreciation.
- ▶ Productivity, terms of trade, and risk premia affect the equilibrium real rate.
- ▶ “Competitive” has no meaning without a price index, horizon, and sector.

Backup: a crawling band trades flexibility for credibility



- The market rate fluctuates inside the corridor while the authorities move its boundaries over time.
- A narrow corridor can strengthen the anchor but invites one-way bets when fundamentals diverge.

Source: Stylized diagram; BCB, 1999 Q1 FX Market Analysis.

Backup: the policy trilemma explains the 1999 choice

Stable exchange rate

Authorities commit resources to limiting currency movements.

Independent monetary policy

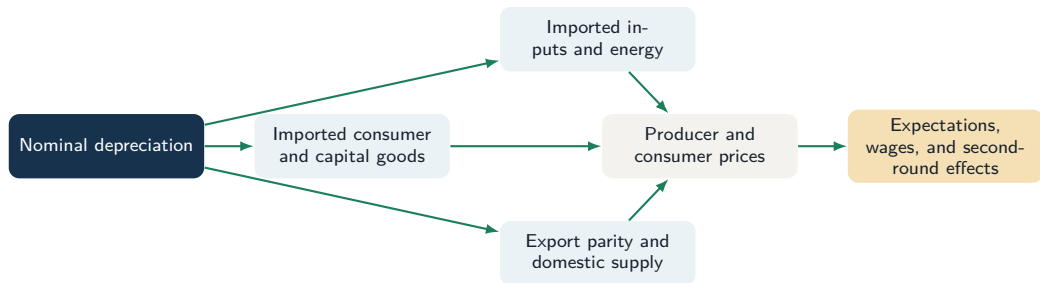
Interest rates respond to domestic inflation and activity.

Capital mobility

Funds can enter or leave in response to returns and risk.

- ▶ A country can fully pursue only two objectives at once.
- ▶ In 1995–98, interest rates defended the exchange rate. Since 1999, the currency floats and monetary policy targets inflation.

Backup: depreciation reaches inflation through several margins



- ▶ A 2026 BCB exercise: a 10% depreciation adds about 1.0 point to consumer inflation over 12 months. The estimate varies across episodes.
- ▶ Monetary policy responds to the inflation forecast, not mechanically to the exchange rate.

Source: BCB, March 2026 Monetary Policy Report, pass-through box; BCB Working Papers 5, 181, and 473.

Backup: reserve accumulation changed Brazil's external buffer



- ▶ Reserves insure against external funding stops and reduce foreign-currency liquidity risk.
- ▶ The stock rose sharply before the global financial crisis and has remained large.

Source: BCB SGS 13621. International reserves, cash concept; last daily observation of each month, US\$ billion.

Backup: monetary policy reacts to the inflation outlook



- ▶ The policy rate is forward-looking; it need not move one-for-one with current inflation.
- ▶ Fiscal risk and exchange-rate pass-through affect how much tightening is required.

Source: BCB SGS 432 and 433. Selic target is the last observation of each month; IPCA is compounded over 12 months.

Backup: full distributional effects of a depreciation

Margin	Possible gain from depreciation	Possible cost
Exporters	Higher domestic-currency revenue	Imported inputs and financing become costlier
Import-competing firms	More price space	Capital goods and technology cost more
Households	Employment may shift toward tradables	Food, fuel, and durable goods become dearer
Public sector	Some revenues rise with nominal activity	Indexed debt and inflation costs can increase
Productivity	Entry into tradables may support learning	Protection can preserve low-productivity firms

The answer depends on balance sheets, imported-input intensity, credibility, and persistence.

Backup: macro scorecard

Area	Achievement	Remaining constraint
Inflation	Hyperinflation ended; expectations have an explicit anchor	Shocks still pass through the currency and indexation
Exchange rate	A float absorbs external shocks	Volatility affects prices, balance sheets, and investment
External sector	Reserves and lower FX mismatch strengthened buffers	Commodity and global-rate cycles remain important
Fiscal policy	Strong revenue capacity and broad social insurance	Rigid spending and debt uncertainty compress investment
Capital formation	Deep domestic financial markets	High cost of capital and low investment

Backup: micro scorecard

Area	Evidence of capability	Binding margin
Technology	Agricultural research and digital finance	Diffusion beyond frontier sectors
Firms	Globally competitive exporters and banks	Long tail of small, informal producers
Tax system	Reform establishes a modern VAT structure	Transition, exemptions, and implementation
Trade	Competitive commodity and selected manufacturing exports	Low integration and costly inputs
Workers	Broad enrollment and recent labor-market strength	Learning gaps and informality
Infrastructure	Scalable concessions and digital public rails	Logistics, sanitation, and local coordination
State capacity	Strong agencies and data systems in selected areas	Uneven local capacity and territorial coercion

Backup: aggregate productivity depends on firms and allocation

$$\text{Aggregate productivity} = \underbrace{\text{productivity within firms}}_{\substack{\text{technology, management,} \\ \text{skills, scale}}} + \underbrace{\text{allocation across firms}}_{\substack{\text{entry, exit, competition,} \\ \text{capital and labor flows}}}$$

Example: if a bakery that produces twice as much per worker wins customers from a less productive one nearby, average productivity rises — even though no single firm changed.

Within-firm margin

Can an existing producer adopt better technology and organization?

Reallocation margin

Do productive firms expand while persistently unproductive firms shrink or exit?

Backup: a long tail of low-productivity firms shapes the aggregate

The long tail

Many small producers use little capital, management technology, or formal finance.

The frontier

A smaller set of firms competes globally and adopts advanced technology.

- ▶ Capabilities decide adoption: targets, inventory control, quality systems, and skills.
- ▶ Complex rules encourage firms to remain small or split activities.
- ▶ Weak exit and reallocation preserve low-productivity uses of capital and labor.

Credit subsidies alone do little when management, competition, and demand remain unchanged.

Source: World Bank, *Brazil: Skills and Jobs* and productivity diagnostics; OECD, *Economic Survey: Brazil 2023*.

Backup: state capacity is an economic input

Administrative capacity

Select projects, hire staff, procure inputs, manage contracts, and use data.

Legal capacity

Enforce contracts, protect rights, resolve disputes, and sanction corruption.

Fiscal capacity

Raise revenue with low compliance costs and allocate it toward high-return services.

Coordination capacity

Align federal, state, and municipal responsibilities across shared networks.

Source: World Bank, The Brazil of the Future; OECD, Government at a Glance: Latin America and the Caribbean.

Backup: spatial inequality changes which reform binds first

Productive structure

Agriculture, industry, services, and public employment have different regional weights.

Local capacity

Municipalities differ in tax bases, technical staff, procurement, and service networks.

Household access

Transport, schools, health, sanitation, and security shape mobility within the same city.

A national reform produces local results through unequal implementation capacity.

Source: IBGE, Regional Accounts and 2022 Population Census; Ipea, regional development studies.

Backup: Brazil is replacing five overlapping taxes with two VATs

2026	2027	2029–32	2033
Test year for the new system.	Federal VAT (CBS) replaces PIS/Cofins.	State-municipal VAT (IBS) phases in for ICMS/ISS.	Full new system.

- ▶ Today, five taxes at three government levels tax the same sale — taxes pile on taxes (“cascading”) and firms litigate for decades.
- ▶ A destination-based VAT taxes only the value each firm adds, reducing production distortions.
- ▶ Federal coordination, exemptions, compliance systems, and transition rules determine the gain: implementation is an economic reform, not an administrative footnote.

Source: Receita Federal, [Consumption Tax Reform](#).

Backup: competition policy changes the return to innovation

When entry is credible

Incumbents invest, lower markups, and adopt technology to defend market share.

When rents are protected

Licensing, procurement design, network bottlenecks, and local political connections can preserve inefficient firms.

- ▶ Antitrust, bankruptcy, trade exposure, and transparent procurement affect the same reallocation margin.
- ▶ The objective is contestability, not a mechanical preference for large or small firms.

Source: OECD, Competition Assessment Reviews: Brazil; CADE, institutional reports.

Backup: trade can raise productivity through inputs, scale, and selection

Channels

- ▶ Cheaper and better intermediate inputs
- ▶ Larger markets for productive exporters
- ▶ Technology and management diffusion
- ▶ Entry and exit under competition

Transition costs

- ▶ Workers and places adjust slowly
- ▶ Logistics and customs can bind first
- ▶ A volatile real exchange rate changes incentives
- ▶ Skills determine who captures the gains

The relevant choice is the sequence and support for adjustment, not autarky versus free trade.

Backup: human capital starts before school and differs by place

Early childhood

Nutrition, health, language exposure, and childcare shape readiness.

School system

Teacher support and instructional time determine whether enrollment becomes learning.

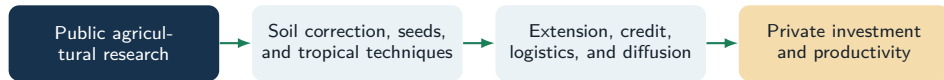
Territory

Transport, violence, sanitation, and local fiscal capacity affect access and continuity.

- ▶ Childcare pays a double dividend: mothers can take formal jobs, and quality care improves children's language, nutrition, and school readiness.
- ▶ Equal spending does not imply equal learning when local constraints differ.

Source: World Bank, Brazil Human Capital Review; IBGE, PNAD Contínua care responsibilities; Inep education indicators.

Backup: Embrapa shows how complementary institutions raise productivity



Lesson

Mission-oriented research works when adoption, finance, and infrastructure reinforce it.

Trade-off

Productivity per hectare and environmental enforcement jointly determine land-use outcomes.

Source: Embrapa Cerrados; Ipea, agricultural productivity studies.

Backup: there is no single organized-crime statistic

Lethal violence

Unit: deaths

National and annual. It does not identify the organization responsible.

Prison intelligence

Unit: groups

SENAPPEN identified at least 88 organizations affecting prisons in its 2024 exercise.

Territorial control

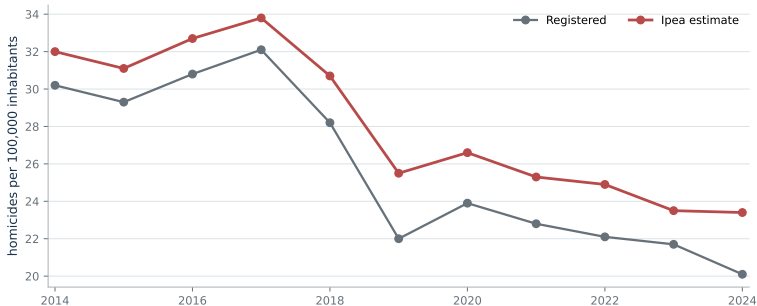
Unit: places and residents

Directly measures armed governance, but only for Metropolitan Rio.

Measurement must match the mechanism being discussed.

Source: Ipea/FBSP, Atlas da Violência 2026; SENAPPEN, Mapa das ORCRIMs; GENI/UFF and Instituto Fogo Cruzado, 2025.

Backup: lethal violence fell, but data quality changes the trend



- ▶ The registered rate fell after 2017; reallocating some deaths of undetermined cause produces a smaller decline.
- ▶ Homicide measures lethal violence, not the size or revenue of criminal organizations.

Source: Ipea and FBSP, Atlas da Violência 2026, based on SIM/Ministry of Health and IBGE. Rates per 100,000. Estimated series is model-based.

Backup: coercive market power reaches ordinary goods and services

Household services

Cooking gas, broadband, cable TV, and electricity.

Mobility

Alternative transport and compulsory payments from drivers.

Property

Irregular construction, forced dispossession, and property sales.

Local retail

Security fees, merchant extortion, and forced suppliers.

Economic wedges

- ▶ illegal tax
- ▶ restricted entry
- ▶ weak property rights
- ▶ distorted procurement

Source: MPRJ cases in Nova Iguaçu (2019), Itaboraí (2019), Queimados (2024), and the illegal-construction task force (2022). These cases do not measure national prevalence.

Backup: full instrument-outcome menu for security policy

Margin	Instruments	Development outcome
Territorial prevention	Safe routes, reliable transport, service continuity, youth prevention	Attendance, business entry, service uptime
Investigation and justice	Forensics, witness protection, case management, corruption control	Clearance with due process, time to disposition
Financial disruption	Beneficial ownership, intelligence, freezing and confiscation	Assets finally recovered, disrupted cash flow
Sector regulation	Internet, fuels, fintech, construction, health, procurement	Prices, entry, ownership transparency
Prison governance	Intelligence, lawful communications control, work and reintegration	Violence, illicit communication, recidivism

Source: MJSP, ENFOC and ENCCLA; SENAPPEN/CNPPC, National Criminal and Penitentiary Policy Plan 2024–2027.

Backup discussion: what outcome would demonstrate greater state capacity?

Choose one instrument and the final outcome that should be used to evaluate it.

Operations

Activity measure

Arrests

Intermediate output

Recovered assets

Financial outcome

Market entry

Development outcome

- ▶ Add one safeguard against abuse or displacement to another territory.
- ▶ Explain why your chosen outcome is closer to welfare than the activity count.

References and data

Data cutoff: July 2026. Series retrieved from official APIs on July 21, 2026; processed CSVs archived with the slides.

Time series (BCB SGS codes)

- ▶ IPCA, monthly: SGS 433
- ▶ Selic target: SGS 432
- ▶ BRL/USD, year-end selling rate: SGS 3692
- ▶ Real GDP growth (IBGE): SGS 7326
- ▶ International reserves, cash: SGS 13621
- ▶ Gross general government debt: SGS 13762
- ▶ Unemployment (PNAD Contínua): SGS 24369
- ▶ Credit spread, outstanding: SGS 20783
- ▶ Convergence: World Bank WDI, NY.GDP.PCAP.KD (constant 2015 US\$)

Reports and studies

- ▶ BCB, Monetary Policy Report, March 2026 (pass-through box); Working Papers 5, 25, 181, 473
- ▶ OECD, Economic Survey: Brazil 2023; PISA 2022 Brazil country note
- ▶ World Bank, The Brazil of the Future (productivity, 1996–2020)
- ▶ IBGE: 2022 Census; Regional Accounts 2021; Population Projections 2024; Desigualdades Sociais por Cor ou Raça
- ▶ Ipea/FBSP, Atlas da Violência 2026
- ▶ GENI/UFF and Instituto Fogo Cruzado, Mapa Histórico dos Grupos Armados 2025
- ▶ UNICEF, Educação Sob Cerco 2025
- ▶ EPE, BEN 2026; MCTI, National Emissions Inventory (2022)
- ▶ Receita Federal, Consumption Tax Reform portal